

**ESLSCA
UNIVERSITY
MBA PROGRAM**

**ACCOUNTING FOR
MANAGERS**

**MANAGERIAL ACCOUNTING
[PART TWO]**

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LECTURE ONE and TWO

COST CLASSIFICATIONS

1. Dickison Corporation reported the following data for the month of December:

Direct Materials.....	\$71,000
Direct Labor cost.....	38,000
Manufacturing Overhead.....	69,000
Selling expenses.....	24,000
Administrative expenses.....	42,000

- a. The product cost= $71000+38000+69000=$
b. The period cost= $24000 +42000=66000$
-

2. A partial listing of costs incurred during December at Gagnier Corporation appears below:

Factory supplies	\$8,000
Administrative wages and salaries	105,000
Direct materials	153,000
Sales staff salaries	68,000
Factory depreciation	49,000
Corporate headquarters building rent	34,000
Indirect Manufacturing labor	32,000
Marketing	103,000
Direct Labor	83,000

- a. The total of the period [non manufacturing] costs listed above for December is:
B. \$89,000 B. 310,000 C. \$325,000 D. \$399,000
- b. The total of the indirect manufacturing overhead costs listed above for December is:
A. \$325,000 B. \$635,000 C. \$89,000 D. \$40,000
- c. The total of the product costs [manufacturing] listed above for December is:
A. \$310,000 B. \$89,000
C. \$635,000 D. \$325,000

3. Lettman Corporation has provided the following partial listing of costs incurred during November:

Marketing Salaries	\$45,000
Property taxes factory	9,000
Administrative travel	98,000
Sales commission	48,000
Indirect labor	38,000
Direct material	165,000
Advertising	138,000
Depreciation of production equipment	39,000
Direct labor	87,000

Required

- What is the total amount of product cost listed above? Show your work.
 $9000 + 38000 + 38000 + 165000 + 39000 + 87000 =$
- What is the total amount of period cost listed above? Show your work.
 $45000 + 98000 + 48000 + 138000 =$

Task Four

Chill Out Novelties sells ice cream bars from a kiosk near campus. Fixed costs are \$400 per week and the variable cost is \$0.50 per ice cream bar. Complete the following table for the levels of ice cream bars sold per week. Round your answers to two decimal places.

Number of ice cream bars	400	800	1,600
Total fixed cost	400	400	400
Fixed cost per bar	1	0.5	0.25
Variable cost per bar	0.5	0.5	0.5
Total variable cost	200	400	800
Total cost	600	800	1200
Total Cost per bar	1.5	1	0.75

Task Five (MCQ)

1- All else being equal, if sales revenue doubles, variable costs will:

- A) decrease in total.
- B) increase in total.**
- C) decrease on a per unit basis.
- D) increase on a per unit basis.

2- When Carter, Inc. sells 48,000 units, its total variable cost is \$115,200. What is its total variable cost when it sells 54,000 units?

$$VC/unit = 115200/48000 = 2.4 \times 54000$$

- | | |
|---------------------|--------------|
| A) \$100,800 | B) \$115,200 |
| C) \$129,600 | D) \$134,800 |

3- Which of the following is a variable cost?

- A) A cost that is \$26,000 when production is 65,000, and \$26,000 when production is 91,000.
- B) A cost that is \$26,000 when production is 65,000, and \$36,400 when production is 91,000.** *VC/unit is the same at both levels of production*
- C) A cost that is \$26,000 when production is 65,000, and \$52,000 when production is 91,000.
- D) A cost that is \$52,000 when production is 65,000, and \$52,000 when production is 91,000.

4- Which of the following statements is true?

- A) Fixed costs are constant on a per unit basis.
- B) Variable costs per unit decrease as activity volume increases.
- C) Variable costs are constant in total dollars.
- D) Fixed costs are constant in total dollars.**

Task Six

Answer the following questions based on the given data:

Sales price.....\$160 per unit

Fixed costs:

Marketing and administrative.....\$20,000 per period

Manufacturing overhead.....\$15,000 per period

Variable Costs:

Marketing\$5 per unit

Manufacturing overhead.....\$30 per unit

Direct labor (manufacturing).....\$10 per unit

Direct materials (manufacturing).....\$40 per unit

Number of units produced and sold during the period.....1000 units

Required:

- How much is the total manufacturing cost COGS ?
- How much is the gross margin (profit)?
- How much is the total variable cost ?
- How much is the total contribution margin ?

<u>Traditional /Absorption</u>		<u>Contribution Approach</u>	
Sales	160,000	Sales	160,000
Less: COGS (a)	95,000	Less: Variable Cost (b)	<u>(85000)</u>
= Gross Profit (d)	65000	Contribution Margin ©	75000
Less: operating exp		Less: Fixed cost	<u>(35000)</u>
Variable 5000			
Fixed 20000	<u>(25000)</u>		
Net Income	<u>40,000</u>	Net Income	<u>40,000</u>

Task Seven

Aspen Inc has the following information for its first year of operations:

Units sold	3,600
Unit sales price	\$ 200.00
Direct material per unit	\$ 60.00
Direct labor per unit	\$ 30.00
Variable manufacturing overhead per unit	\$ 30.00
Fixed manufacturing overhead	\$ 115,200
Variable selling expenses	\$ 15.00
Fixed selling and administrative expenses	\$ 65,000

- a. Calculate total variable cost and total fixed costs .

Total VC

$$60 \times 3600 + 30 \times 3600 + 30 \times 3600 + 15 \times 3600 =$$

$$\text{TFC} = 115200 + 65000 = 180200$$

- b. Prepare contribution approach income statement.

Sales	$200 \times 3600 = 720000$
-TVC	$135 \times 3600 = (486000)$
=TCM	$65 \times 3600 = 234000$
-TFC	(180200)
=Operating profit	53800

Task Eight

Whitman Corporation, a merchandising company, reported sales of 7,400 units for May at a selling price of \$677 per unit. The cost of goods sold (all variable) was \$441 per unit and the variable selling expense was \$54 per unit. The total fixed selling expense was \$155,600. The variable administrative expense was \$24 per unit and the total fixed administrative expense was \$370,400.

Required:

- a. Prepare a contribution format income statement for May.
- b. Prepare a traditional format income statement for May.

Sales (7,400 units × \$677 per unit)		\$5,009,800
Variable expenses:		
Cost of goods sold (7,400 units × \$441 per unit)	\$3,263,400	
Variable selling expense (7,400 units × \$54 per unit)	399,600	
Variable administrative expense (7,400 units × \$24 per unit)	177,600	3,840,600
Contribution margin		1,169,200
Fixed expenses:		
Fixed selling expense	155,600	
Fixed administrative expense	370,400	526,000
Net operating income		<u>\$643,200</u>

Task Nine

Stewart Company has no beginning and ending inventories, and reports the following information about its only product:

Direct materials used	\$29,000
Direct labor	\$17,000
Variable indirect production	\$13,000
Fixed indirect production	\$18,000
Variable selling and administrative expenses	\$22,000
Fixed selling and administrative expenses	\$11,000
Units produced and sold	10,000
Selling price per unit	\$25

Required:

- A) Prepare an income statement using the contribution approach.

$$\text{Sales } 250000 - \text{TVC } 81000 = \text{TCM } 169000 - \text{TFC } 29000 = \text{NI } 140000$$

- B) Prepare an income statement using the traditional approach.

$$\text{Sales } 250000 - \text{COGS } 77000 = \text{Gross Profit } 173000 - 33000 = \text{NI } 140000$$

Task Ten

Honey Corporation, a merchandising company, reported the following results for January:

Number of units produced	5,800
Selling price per unit	\$892
Units costs of goods sold	\$517
Variable selling expense per unit	\$31
Total fixed selling expense	\$152,600
Variable administrative expense per unit	\$48
Total fixed administrative expenses	\$390,200

Cost of goods sold is a variable cost in this company.

Required:

- A) Prepare a traditional format income statement for January.

$$\text{Sales} - \text{Manufacturing costs} = \text{Gross profit} - \text{Selling and operating cost} = \text{net income}$$

- B) Prepare a contribution format income statement for January.

$$\text{Sales} - \text{Total variable costs} = \text{Contribution margin} - \text{total fixed costs} = \text{net income}$$

a. Prepare a traditional format income statement for January.

Sales – Manufacturing costs = Gross profit – Selling and operating cost = net income

<i>Sales</i>	<i>5800 x 892 =</i>	<i>5173600</i>
<i>– Manufacturing costs (5800 x 517) =</i>	<i>2998600</i>	
<i>= Gross profit</i>		<i>2175000</i>
<i>– Selling (31 x 5800) + 152600 =</i>	<i>(332400)</i>	
<i>and administrative cost</i>		
<i>(48 x 5800 + 390200)</i>	<i>(668600)</i>	
<i>= net income</i>		<i>1174000</i>

b. Prepare a contribution format income statement for January.

Sales – Total variable costs = Contribution margin – total fixed costs = net income

<i>Sales</i>	<i>5173600</i>
<i>- Total variable cost</i>	
<i>(2998600 + 31 x 5800 + 48 x 5800)</i>	<i><u>(3456800)</u></i>
<i>Contribution margin</i>	<i>1716800</i>
<i>– total fixed costs 152600 + 390200</i>	<i><u>(542800)</u></i>
<i>= net income</i>	<i>1174000</i>

LECTURE TWO: BRAK EVEN ANALYSIS (CVP Analysis)

TASK ONE

The following information describes a product expected to be produced and sold by Garrison Company:

Selling price	\$80 per unit
Variable costs	\$32 per unit
Total fixed costs	\$630,000

Required:

- (a) Calculate the contribution margin ratio.
- (b) Calculate the break-even point in dollar sales.
- (c) What dollar amount of sales would be necessary to achieve a pretax income of \$120,000?

Answer:

- (a) $\text{Contribution margin ratio} = (\$80 - \$32) / \$80 = 60\%$
- (b) $\text{Break-even point in dollars} = \$630,000 / 0.6 = \$1,050,000$
- (c) $\text{Pretax income of } \$120,000 = (\$630,000 + \$120,000) / 0.6 = \$1,250,000$

TASK TWO

Bing Company's contribution margin income statement is presented below. Sales for the current period consisted of 7,500 units.

Bing Company Contribution Margin Income Statement	
Sales	\$225,000
Variable costs	<u>135,000</u>
Contribution margin	90,000
Fixed costs	<u>48,000</u>
Net income	<u>\$42,000</u>

Required:

- Q1: Compute the company's break-even point in
- (a) units, and
 - (b) dollars.

Q2: Compute the margin of safety in
(b) dollars and (d) percent.

Answer:

Per unit costs:

$$\text{Sales price} = \$225,000/7,500 = \$30$$

$$\text{Variable cost} = \$135,000/7,500 = \$18$$

$$\text{Contribution margin per unit} = \$30 - \$18 = \$12 \text{ per unit}$$

$$\text{Contribution margin ratio} = (\$30 - \$18)/\$30 = .40$$

$$(a) \text{ Break-even in units} = \$48,000/\$12 = 4,000 \text{ units}$$

$$(b) \text{ Break-even in dollars} = \$48,000/.40 = \$120,000 \text{ (or } 4,000 \text{ units} \times \$30)$$

$$(c) \text{ Margin of safety in dollars} = \$225,000 - \$120,000 = \$105,000$$

$$(d) \text{ Margin of safety in percent} = \$105,000/\$225,000 = 46.7\%$$

TASK THREE

Longiotti Corporation produces and sells a single product. Data concerning that product appear below:

Selling price per unit\$150

Variable expense per unit.....36

Fixed expense per month.....\$159,600

Required:

Determine the monthly break-even in total dollar sales.

Answer:

$$CM/unit = 150 - 36 = 114$$

$$BEP \text{ in units} = 159600/114 = 1400 \text{ units} \times 150 = 210,000$$

$$Cm\% = 114/150 = 0.76$$

$$BE \$ = 159600/0.76 = 210,000$$

TASK FOUR

Magnolia Company is considering the production and sale of a new product with the following sales and cost data: unit sales price, \$350; unit variable costs, \$180; total fixed costs, \$399,500; and projected sales, \$910,000. Round your answers to the nearest whole unit or dollar.

(a) Calculate break-even in units.

(b) Calculate break-even in dollars (use four decimal places when calculating the contribution margin ratio).

- (c) Calculate number of units that would need to be sold to generate an after-tax profit of \$420,000 assuming a 30% tax rate.
- (d) Calculate dollar sales that would be needed to generate the same profit as above.
- (e) Calculate the margin of safety stated as a percentage using the \$910,000 projected sales level.

EBT 100	tax 30	Net income 70
x		420000

Answer:

(a) $\$399,500/(\$350-180) = 2,350 \text{ units}$

(b) $CM \text{ ratio} = 350-180/350 = 0.4857$

$\$399,500/0.4857 = \$822,524 \text{ or } \$822,500.$

(c) $\$399,500 + (\$420,000/(0.7)) = (\$399,500 + \$600,000)/\$170 = 5,880 \text{ units}$

(d) $(\$399,500 + \$600,000)/.4857 = \$2,057,855 \text{ or } \$2,058,000 \text{ or } (5,880 \text{ BE units} \times \$350 = \$2,058,000)$

(e) $\$910,000 - \$822,500 = \$87,500/\$910,000 = 9.6\%$

TASK FIVE

Iron Decor manufactures decorative iron railings. In preparing for next year's operations, management has developed the following estimates:

	Total	Per Unit
Sales (20,000 units).....	\$1,000,000	\$50.00
Direct materials.....	\$200,000	\$10.00
Direct labor (variable).....	\$50,000	\$2.50
Manufacturing overhead:		
Variable.....	\$70,000	\$3.50
Fixed.....	\$80,000	\$4.00
Selling & administrative:		
Variable.....	\$100,000	\$5.00
Fixed.....	\$30,000	\$1.50

Answer:

a. *Unit contribution margin.* $SP-VC/unit = 50 - 10 - 2.5 - 3.5 = 29$

b. *Contribution margin ratio.* $CM / SP = 29/50 = 0.58 = 58\%$

c. *Break-even in dollar sales.* $TFC / CM\% = 110,000 / 0.58 = \$189,655$

$BEP \text{ units} = 110,000 / 29 = 3793$ $BE\$ 3793 \times 50 = \$189,655$

d. *Margin of safety percentage.*

$MOS = \text{Actual sales} - BEP = 20,000 - 3793 = 16,207 \text{ units} / 20,000 = 81\%$

$\text{OR } 1,000,000 - 189,655 = 810,345 \text{ MOS}\% = 810,345 / 1,000,000 = 81\%$

TASK SIX:

Winston Company has variable costs of \$5 per unit and a selling price of \$10 per unit. Fixed costs are \$100,000. Planned unit sales for 2015 are 25,000 units. Actual unit sales for 2014 were 22,000 units. What is the margin of safety in units for 2015?

A) 2,000 units

B) 3,000 units

C) 5,000 units

D) 7,000 units

Answer: $BEP\ in\ units = 100,000 / 5 = 20,000\ units$

$MOS\ in\ units = 25,000 - 20,000 = 5000\ (C)$

TASK SEVEN

If the selling price per unit \$100

Variable costs per unit \$80

Total fixed costs \$80,000

If fixed costs increased by 10% and management wanted to maintain the original break-even point, then the selling price per unit would have to be increased to

A) \$101.00

B) \$102.40

C) \$102.00

D) \$103.00

Answer:

$$BEP\ in\ units = Total\ fixed\ costs / SP - VC$$
$$= 80,000 / 20 = 4000\ units$$

$$Then\ 4000 = 88000 / SP - 80$$

$$(SP - 80) = 88,000 / 4,000$$

$$SP - 80 = 22 \qquad SP = 22 + 88 = 102$$

TASK EIGHT

Nocum Corporation has provided the following contribution format income statement. Assume that the following information is within the relevant range.

Sales (3,000 units)	\$ 120,000
-Variable expenses	<u>90,000</u>
=Contribution margin	30,000
-Fixed expenses	<u>21,000</u>
=Net operating income	<u>\$ 9,000</u>

If sales decline to 2,900 units, the net operating income would be closest to:
 $9000 - (10 \times 100 \text{ units}) = \8000

TASK NINE

Duve Corporation has provided the following contribution format income statement. Assume that the following information is within the relevant range.

Sales (2,000 units)	\$ 40,000
Variable expenses	<u>24,000</u>
Contribution margin	16,000
Fixed expenses	<u>11,200</u>
Net operating income	<u>\$ 4,800</u>

If the selling price increases by \$4 per unit and the sales volume decreases by 200 units, the net operating income would be closest to:

Current contribution margin $= 16000 / 2000 = 8$

New CM/unit = 12 Units $2000 - 200 = 1800$

Net operating profit = New CM $= 12 \times 1800 - \text{TFC } 11200 = 10400$

TASK TEN

Chilton, Inc. sold 11,000 units last year for \$20 each. Variable costs per unit were \$4 for direct materials, \$1.50 for direct labor, and \$2.50 for variable overhead. Fixed costs were \$60,000 in manufacturing overhead and \$40,000 in non-manufacturing costs.

- a. What is the unit contribution margin?

$$20 - (4 + 1.5 + 2.5) = 12$$

- b. What is the total contribution margin?

$$12 \times 11000 = 132000$$

- c-. If sales increase by 2,000 units, by how much will profits increase?

$$\text{Profits will increase by the additional CM} = 12 \times 2000 = 24000$$

TASK ELEVEN

Vaughn, Inc. sold 17,000 units last year for \$50 each. Variable costs per unit were \$15 for direct materials, \$15 for direct labor, and \$10 for variable overhead. Fixed costs were \$10,000 in manufacturing overhead and \$50,000 in nonmanufacturing costs.

- a. What is the unit contribution margin?

$$50 - (15 + 15 + 10) = 10$$

- b. What is the total contribution margin?

$$10 \times 17000 = 170,000$$

- c. What is the contribution margin ratio?

$$10/50 = 20\%$$

- d. If sales increase by 5,000 units, by how much will profits increase?

$$5000 \times 10 = 50000$$

TASK TWELVE

Stanger Inc. produces and sells two products. Data concerning those products for the most recent month appear below:

	Product N	Product X
Sales	\$14,000	\$27,000
Variable expenses	\$ 6,720	\$10,550

Fixed costs for the entire company are \$17570

Required

- Determine the overall break-even point for the company. Show your work!
- If the sales mix shifts toward Product N with no change in total sales, what will happen to the break-even point for the company? Explain.

Answer:

	<i>Product N16S</i>	<i>Product X07D</i>	<i>Total</i>
<i>Sales</i>	<i>\$14,000 (34.1%)</i>	<i>\$27,000 (65.9%)</i>	<i>41000</i>
<i>Variable expenses</i>	<i>\$6,720</i>	<i>\$12,550</i>	<i>19270</i>
<i>CM</i>	<i>7280 CM 52%</i>	<i>14450 CM 53.5%</i>	<i>21730</i>
<i>TFC</i>			<i>17570</i>

% of product N in sales mix = 14000/41000 = 34.1%

% of product Z = 27000 /41000 = 65.9%

WA of CM% = 21730 /41000 = 0.53

BEP = 17570 /0.53= \$33151

BEP N = 33151 x 34.1%= 11304

BEP X = 33151 x 65.9%= 21847

If the sales mix shifted towards product N16S (the lower CM%) the BEP will increase

TASK THIRTEEN

The following monthly data in contribution format are available for the MN Company and its only product, Product SD:

	Total	Per Unit
Sales.....	\$83,700	\$279
Variable expenses	<u>32,700</u>	<u>109</u>
Contribution margin.....	<u>51,000</u>	<u>\$170</u>
Fixed expenses	<u>40,000</u>	
Net operating income.....	<u>\$11,000</u>	

The company produced and sold 300 units during the month and had no beginning or ending inventories.

Required:

a-Without resorting to calculations, what is the total contribution margin at the break-even point? *Answer: 40000*

b. Management is contemplating the use of plastic gearing rather than metal gearing in Product SD. This change would reduce **variable expenses by \$18** per unit. The company's sales manager predicts that this would reduce the overall quality of the product and thus would result in a decline in sales to a level of 250 units per month. Should this change be made? New VC =91

Answer:

$Sales\ 250 \times 279 = 69750 - VC\ 250 \times 91 = 22750 = CM\ 47000 - 40000 = NI = 7000$

Reject

c. Assume that MN Company is currently selling 300 units of Product SD per month. Management wants to increase sales and feels this can be done by cutting the selling price by \$22 per unit and increasing the advertising budget by \$20,000 per month. Management believes that these actions will increase unit sales by 50 percent. Should these changes be made?

Answer:

$Sales\ (300 \times 150\%) = 450 \text{ units} \times 257 = 115650 - (450 \times 109) = CM\ 66,600 - 60000 = 6600 < \text{current income } 11000$

Reject because income will decline by \$4400

d. Assume that MN Company is currently selling 300 units of Product SD. Management wants to automate a portion of the production process for Product SD. The new equipment would reduce direct labor costs by \$20 per unit but would result in a monthly rental cost for the new robotic equipment of \$10,000. Management believes that the new equipment will increase the reliability of Product SD thus resulting in an increase in monthly sales of 12%. Should these changes be made?

Answer:

$\text{New Contribution Margin} = 300 \times 112\% = [336 \times 190] - \text{Fixed costs } 50000 = 13840$

Accept

TASK 14

Morse Company reports total contribution margin of \$48,000 and pretax net income of \$12,000 for the current month. The degree of operating leverage is:

- A) 4.0
- B) 0.25
- C) 1.25
- D) 2.5
- E) 250%

Answer: $DOL = 48000/12000 = 4$

TASK 15

Fielder Productions reports the following information:

Total contribution margin.....	\$32,000
Total fixed costs.....	\$28,000

Required:

- a. Calculate Fielder's degree of operating leverage (DOL).
- b. If sales increase by 6%, what is the expected percentage increase in pretax income?

Answer:

- a. Net Income = $32000 - 28000 = 4000$
- b. $DOL = 32000/4000 = 8$
- c. Percentage increase in income = $8 \times 6\% = 48\%$

TASK 16.

A company is looking into two alternative methods of producing its product. The following information about the two alternatives is available. If the company's expected sales volume is 35,000 units, which alternative should be selected?

Required:

1-Prepare forecasted contribution margin income statements and compute the degree of operating leverage to assess the alternatives.

	<u>Alternative #1</u>	<u>Alternative #2</u>
Variable costs per unit.....	\$8	\$12
Fixed costs.....	\$240,000	\$140,000
Selling price per unit.....	\$20	\$20

2- Calculate the indifferent point for the two alternatives

3-Assume that the company is expecting to sell 22000 units, Prepare forecasted contribution margin income statements and compute the degree of operating leverage to assess the alternatives.

	<u>Alternative #1</u>	<u>Alternative #2</u>
Contribution Margin	12 x 35000=420000	8 x 35000=280,000
Net Income	180,000	140000
DOL = CM/Net Income	2.3	2
CHOOSE Alternative 1		

Indifferent Point

$$\begin{array}{lcl} \text{TC alt \#1} & = & \text{TC alt \#2} \\ 240,000 + 8Q & = & 140,000 + 12Q \\ & & Q = 25000 \end{array}$$